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Case Study



Platform Partner unstop

Case Study 2: Thermal Management – Innovative Solutions for an Energy-Efficient Future

I. Background

Rising global temperatures, increasing urbanization, and growing energy consumption are creating a challenge for sustainable development. Cooling and temperature control now account for a major portion of electricity consumption in residential, commercial, and industrial sectors. The demand for thermal management in buildings, transportation systems, electronic devices, data centres, and energy storage systems continues to grow, resulting in increased energy costs and carbon emissions. To address these challenges, researchers and industries worldwide are exploring innovative thermal management technologies that can reduce heat gain, improve energy efficiency, and support sustainability goals.

II. The Opportunity

Imagine a future where buildings require minimal air conditioning, vehicles remain cooler under direct sunlight, data centres consume less energy for cooling, and heat islands are significantly reduced. The next generation of thermal management solutions may originate from advanced coatings and surface technologies, smart materials and adaptive systems, solar heat reflection systems, building-integrated cooling systems, glazing systems, smart windows, passive cooling technologies, reflective surfaces, radiative cooling systems, thermal energy storage materials, phase change materials, insulation technologies, novel architectural designs and nature-inspired thermal regulation concepts.

Participants are encouraged to think beyond conventional solutions and explore innovative technologies from any scientific or engineering domain that has commercial potential.

III. Key Challenges to Address

An effective idea or solution should:

- Deliver measurable energy savings
- Be economically viable
- Be scalable for large-scale adoption
- Should perform under diverse environmental conditions
- Contribute to sustainability and carbon reduction goals
- Create a competitive advantage through technological innovation

IV. The scope of the case study

Develop an innovative and scalable thermal management solution capable of reducing energy consumption, improving thermal comfort, or enhancing temperature control in one or more of the following application areas:

- Buildings and architectural facades, windows and glass systems, roofs and exterior surfaces, transportation (automobiles, railways, aerospace), consumer products, data centres, EV battery systems, industrial infrastructure, urban environments.
- The solution may involve material innovation, product innovation, process innovation and/or design innovation.
- Do not restrict your ideas to solutions from coatings industry only. Innovative and original ideas will be scored higher and will have a greater chance to be selected for the next round. Simply reproducing an AI generated idea without original thinking will greatly reduce the chance of qualification. Wherever any content is taken or generated via AI, it should be clearly mentioned.
- While a proof of concept or prototype is preferred and will be given extra points, extensive reading and coming up with an innovative idea or solution from a different industry or different areas of science will also enable the selection to the next round.

V. Questionnaire

The proposed solution should demonstrate clear technical differentiation and a measurable impact on energy savings. Participants should address the following questions:

- What specific thermal management challenge is being addressed? Define your target application area and solution.
- Describe the proposed technology or concept.
- What makes the solution unique?
- Explain the scientific or engineering mechanism. How does the solution maintain temperature neutrality or save energy?
- Give a perspective on the handling of the raw materials used and the scalability of the idea. How will the solution be produced?
- How will it be applied or integrated into the target application?
- Define measurable success criteria such as ambient or surface temperature reduction, solar heat reflection, energy savings. Share test protocol and details on system durability.
- Give a perspective on cost (although cost will not be a barrier for the idea selection).

References

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- Levinson, R., Akbari, H., & Berdahl, P. Cool Roofs and Pavements: Increasing Solar Reflectance to Reduce Urban Heat Islands. Energy and Buildings.
- Ahmed A. Alkrush, Mohamed S. Salem, O. Abdelrehim, A.A. Hegazi, Data centers cooling: A critical review of techniques, challenges, and energy saving solutions, International Journal of Refrigeration, Volume 160, 2024, Pages 246-262.